Project Report: Medical Diagnosis Expert System

VITyarthi Project (2026)

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This report provides a comprehensive overview of the **Medical Diagnosis Expert System**, a hybrid AI project that combines rule-based logic with machine learning to assist in preliminary disease identification.

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1. Project Overview

The Medical Diagnosis Expert System is an AI-driven tool designed to bridge the gap between traditional expert systems and modern probabilistic modelling. By using a hybrid approach, the system minimizes the "brittleness" of logic-only rules while maintaining the explainability that pure machine learning models often lack. It is structured to handle ten common diseases by analyzing patient-reported symptoms.

2. Prerequisites

To run or develop this system, the following environment setup is required:

- **Python 3.8+**: The core programming language.
- **Libraries**:
  - scikit-learn: For the Multinomial Naive Bayes implementation.
  - numpy: For numerical operations and feature vector handling.
- **Knowledge Base**: A curated dictionary of diseases containing required symptoms, optional symptoms, severity levels, and medical advice.

3. Core Architecture & Methodology

The system follows a modular architecture as seen in the provided documentation, splitting the task into four distinct layers.

Knowledge Representation

Medical knowledge is encoded as a structured dictionary. This simulates a medical textbook, where each disease is defined not just by a list of symptoms, but by their priority (Required vs. Optional). This distinction allows the system to distinguish between a "perfect match" and a "probabilistic possibility."

Rule-Based Inference Engine

This module utilizes **Forward Chaining** logic. The system checks if the patient's input contains 100% of the "Required" symptoms for a specific disease.

- **If the rule fires:** It calculates a confidence score based on how many "Optional" symptoms are also present.
- **If a required symptom is missing:** The rule does not fire, preventing false positives for diseases that have non-negotiable symptoms (e.g., a "severe headache" is a prerequisite for a Migraine).

Machine Learning Classifier

The system uses a **Multinomial Naive Bayes** model. Since real-world medical data is often messy, the system "auto-trains" itself by generating synthetic samples from the Knowledge Base.

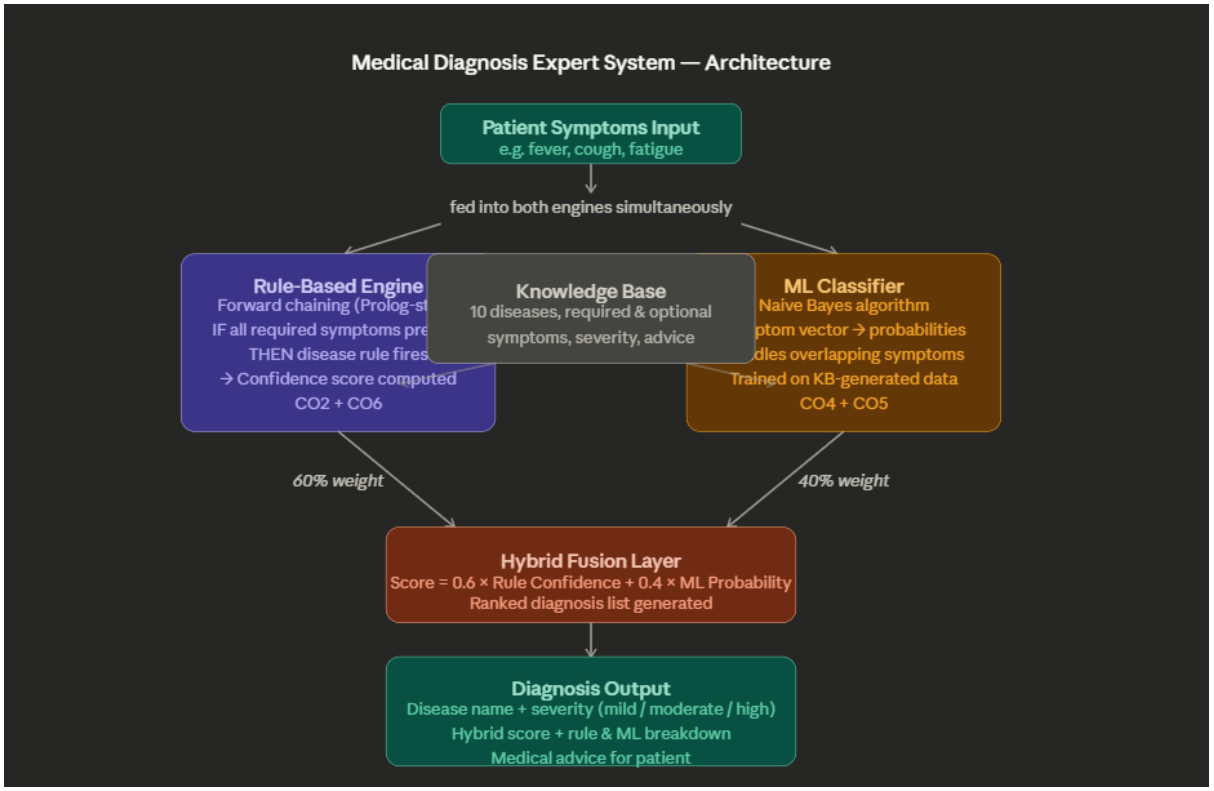
- **Feature Engineering:** Symptoms are converted into a binary vector (1 if present, 0 if absent).
- **Probability:** Unlike the rule engine, the ML classifier provides a probability distribution across all diseases, allowing it to suggest matches even if a patient forgets to mention a "required" symptom.

Hybrid Fusion Layer

The final output is determined by a weighted formula:

$$Final\ Score\ =\ (0.6\ \times RuleConfidence) + (0.4 \times MLProbability)$$

This weighting ensures that logical, rule-based certainty takes precedence while the ML model provides a "safety net" for borderline cases.



4. System Features

- **Severity Categorization:** Diagnoses are tagged as Mild (🟢), Moderate (🟡), or High Risk (🔴).
- **Actionable Advice:** Every diagnosis includes immediate "next steps," such as hydration or seeking emergency care.
- **Explainability:** The system explains its reasoning, showing exactly which symptoms matched the rules and which were predicted by the ML model.

5. Conclusion

This project successfully demonstrates that hybrid AI systems are more robust than single-method approaches. By combining the precision of **Knowledge Representation (CO2)** and **Logical Inference (CO6)** with the flexibility of **Machine Learning (CO4)**, the expert system provides a reliable framework for preliminary medical screening.

6. Future Improvements

- **Deep Learning Integration:** Replacing Naive Bayes with a Neural Network for more complex pattern recognition.
- **Natural Language Processing (NLP):** Allowing users to describe symptoms in plain English rather than selecting from a list.
- **User Profiles:** Storing patient history to track symptom progression over time.
- **API Expansion:** Connecting the system to real-time medical databases (like ICD-10) for a wider range of disease coverage.

7. Reference Links

- **Scikit-Learn Documentation:** https://scikit-learn.org/stable/
- **Forward Chaining in AI:** https://www.geeksforgeeks.org/forward-chaining-and-backward-chaining-in-ai/
- **Naive Bayes Explained:** https://towardsdatascience.com/naive-bayes-classifier-explained-50f9723c3165